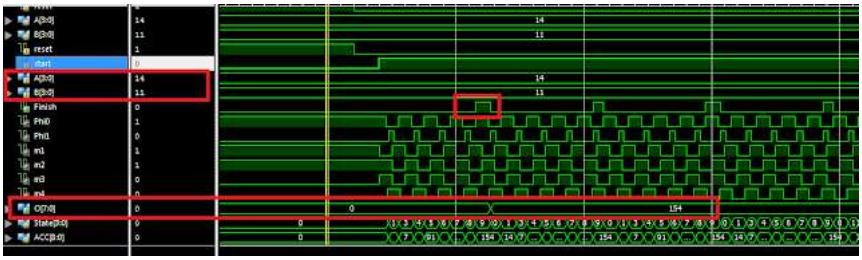
Verilog code for 4x4 Multiplier

This project is to implement a 4x4 multiplier using Verilog HDL. Full Verilog code for the multiplier is presented.
The technique being used is shift/add algorithm, but the different feature is using a two-phase self-clocking system in order to reduce the multiplying time by half.



Verilog code for the multiplier:

```
`timescale 1ns / 1ps
// fpga4student.com FPGA projects, Verilog projects, VHDL projects
// multiplier 4x4 using Shift/Add Algorithm and 2-phase clocking system
// Verilog project: Verilog code for multiplier
module mult_4x4(
         input reset,start,
         input[3:0] A,B,
         output [7:0] O, output Finish
             );
reg [7:0] O;
wire Finish;
wire Phi0,Phi1;// 2 phase clocking
wire m1,m2,m3,m4;
// state machine
reg[3:0] State;
// Accumulator
reg [8:0] ACC; // Accumulator
// logic to create 2 phase clocking when starting
nand u0(m1,start,m2);
buf #20 u1(m2,m1);
buf #10 u2(Phi0,m1);// First phase clocking
not #2 u5(m4,Phi0);
assign m3=~m1;
and #2 u4(Phi1,m3,m4);// Second phase clocking
assign Finish = (State==9)? 1'b1:1'b0; // Finish Flag
// FSM
always @(posedge Phi0 or posedge Phi1 or posedge reset)
begin
 if(reset) begin
 State <= 0;
 ACC <= 0;
 O <= 0;
 end
 else if((Phi0==1'b1) || (Phi1==1'b1)) begin // 2 phase clocking
 if(State==0)
 begin
 ACC[8:4] <= 5'b00000; // begin cycle
 ACC[3:0] <= A; // Load A
 State <= 1;
 end
 else if(State==1 || State == 3 || State ==5 || State ==7)
                // add/shift State
 begin
```

```
 if(ACC[0] == 1'b1) begin // add multiplicand
 ACC[8:4] <= {1'b0,ACC[7:4]} + B;
 State <= State + 1;
 end
 else
 begin
 ACC <= {1'b0,ACC[8:1]};// shift right
 State <= State + 2;
 end
 end
 else if(State==2 || State == 4 || State ==6 || State ==8)
                // shift State
 begin
 ACC <= {1'b0,ACC[8:1]}; // shift right
 State <= State + 1;
 end
 else if(State == 9) begin
 State <= 0;
 O <= ACC[7:0];
 end
 end
end

endmodule
// TestBench
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// Verilog project: Verilog code for multiplier


module test();
// signals
reg start,reset;
reg[3:0] A,B;
// Outputs
wire [7:0] O;
wire Finish;
// device under test
mult_4x4 dut(reset,start, A,B,O,Finish);
initial begin
reset=1; // reset
#40 start = 0;A =14; B= 11;
#400 reset = 0;
#40 start = 1; // start
//@(posedge Finish);
//start = 0;
//$finish;
end

endmodule
```

Simulation result for the 4x4 multiplier:

As soon as the start signal is asserted, the multiplier begins to perform the multiplication. By creating 2 phase clocks, it reduces multiplying time by half.
The finish signal to inform the multiplier that the multiplication has been done and the result is ready.

Simulation result verified the correct operation of the multiplier which 14 multiplies by 11 getting 154 at the output of the multiplier.

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